Solution Architecture — BP Internet Banking

C4 Design · Internet Banking System

> *Document following the C4 model (Context → Container → Component) with architectural justifications per exercise requirements.*

---

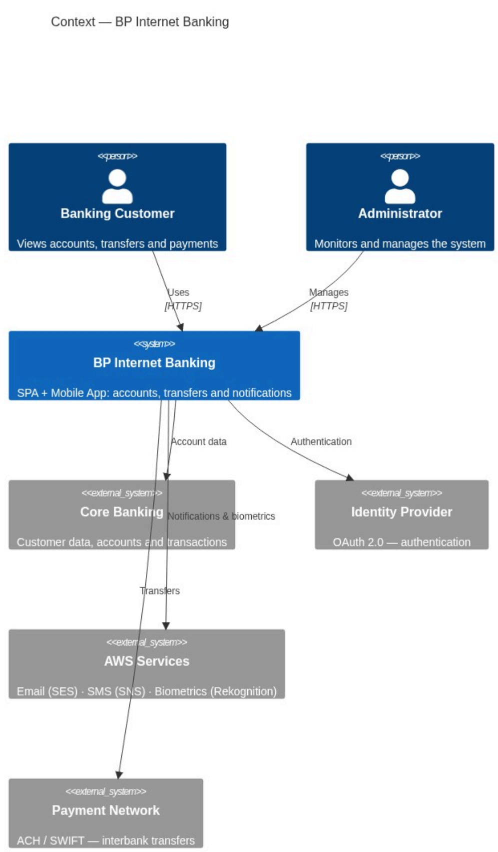
1. Executive Summary

BP requires an internet banking system for account viewing, own and interbank transfers, and real-time notifications. The proposed solution is a **microservices architecture on AWS** with two front-end channels (SPA + Mobile App), OAuth 2.0 + PKCE authentication, biometric onboarding, a centralized integration layer, append-only audit logging, and multi-region high availability.

---

2. Context Diagram (C4 — Level 1)

> *For non-technical audiences. Shows BP Internet Banking and its relationships with users and external systems.*



Actors and External Systems

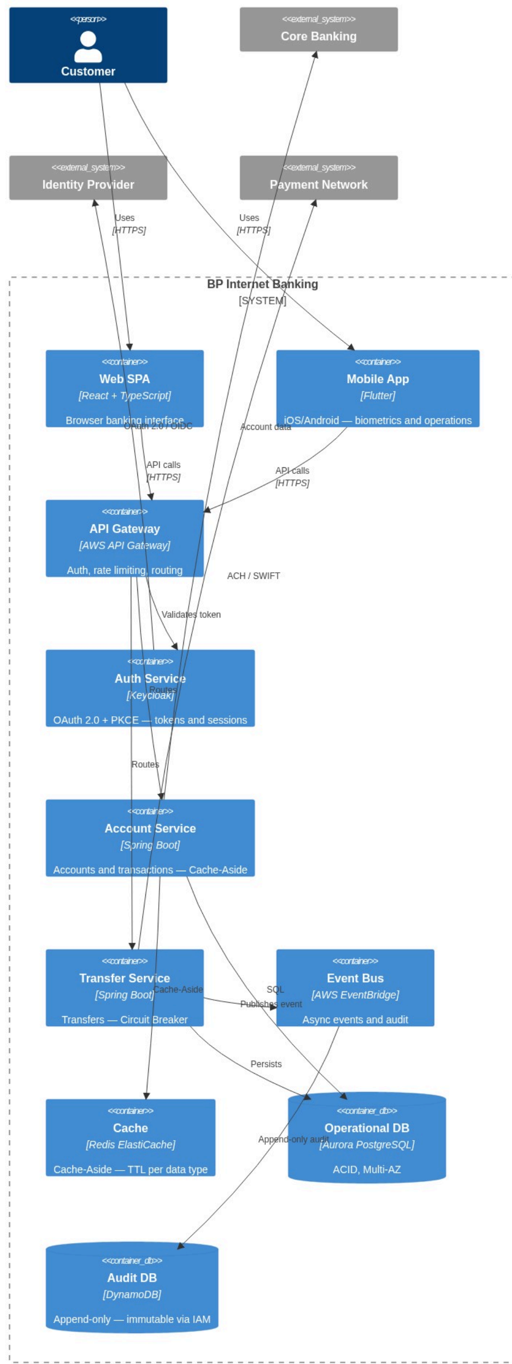
Actor / System	Type	Role
Banking Customer	Person	Retail user — accounts, history, transfers
Operations Administrator	Person	Internal staff — monitoring and alerts
Core Banking Platform	External system	Primary source: customers, transactions, products, profile
Identity Provider (OAuth 2.0)	External system	Issues and validates tokens
AWS Services (SES · SNS · Rekognition)	External system	Email alerts, SMS/OTP, facial verification
ACH/SWIFT Payment Network	External system	Interbank transfer execution

---

3. Container Diagram (C4 — Level 2)

> *For technical audiences. Shows applications, services, databases and messaging with technologies and protocols.*

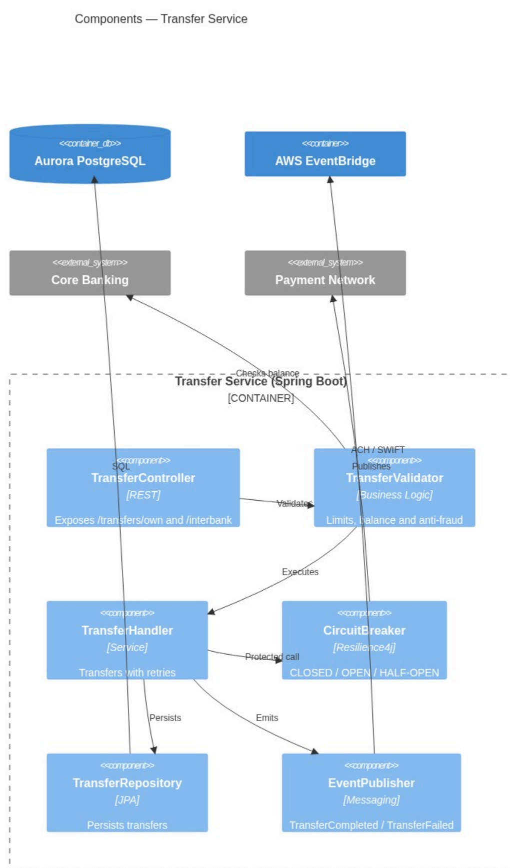
Containers — BP Internet Banking



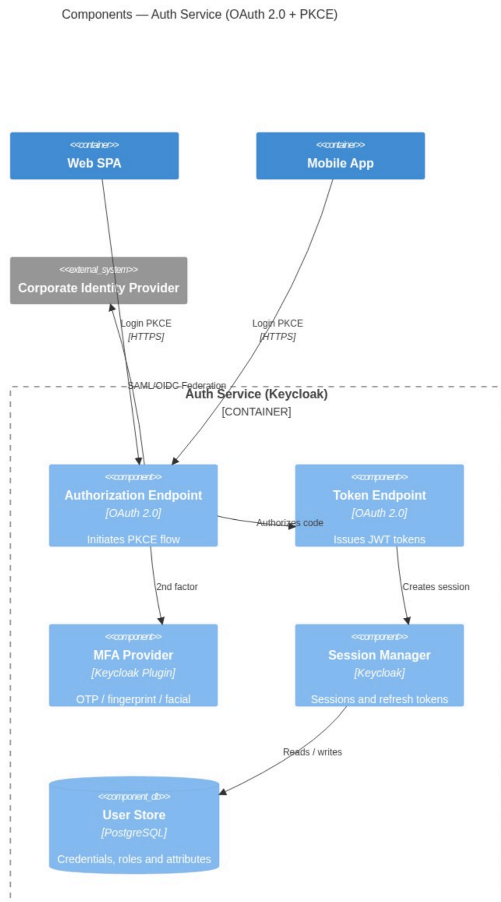
4. Component Diagrams (C4 — Level 3)

4.1 Transfer Service — Components

> *Critical service: handles own and interbank transfers with Circuit Breaker fault tolerance.*

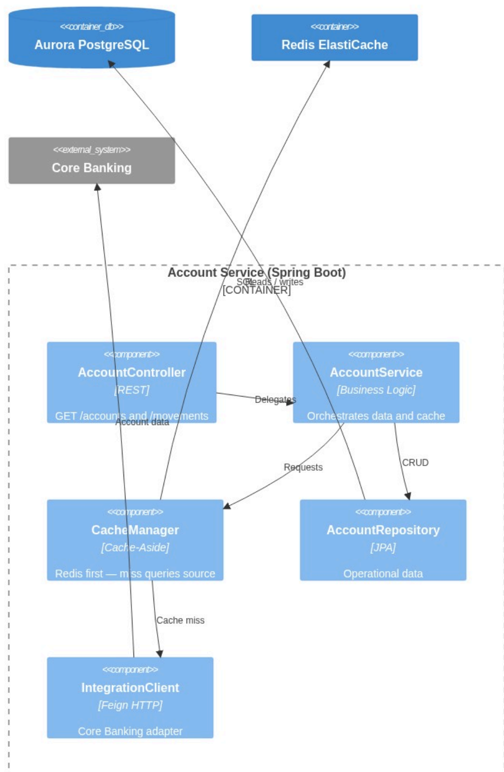


4.2 Auth Service — Components (Keycloak + PKCE)



4.3 Account Service — Components (Cache-Aside)

Components — Account Service (Cache-Aside)



5. Authentication Architecture — OAuth 2.0 + PKCE

Recommended Flow: Authorization Code Flow + PKCE

Why Authorization Code + PKCE and not Implicit Flow?

Criterion	Implicit Flow	Authorization Code + PKCE
Token exposure	Access token in URL (insecure)	Only temporary code in URL
Mobile security	Not recommended (RFC 8252)	Recommended standard
Refresh token	Not supported	Yes, with rotation
Standard status	**Deprecated (OAuth 2.1)**	**Actively recommended**

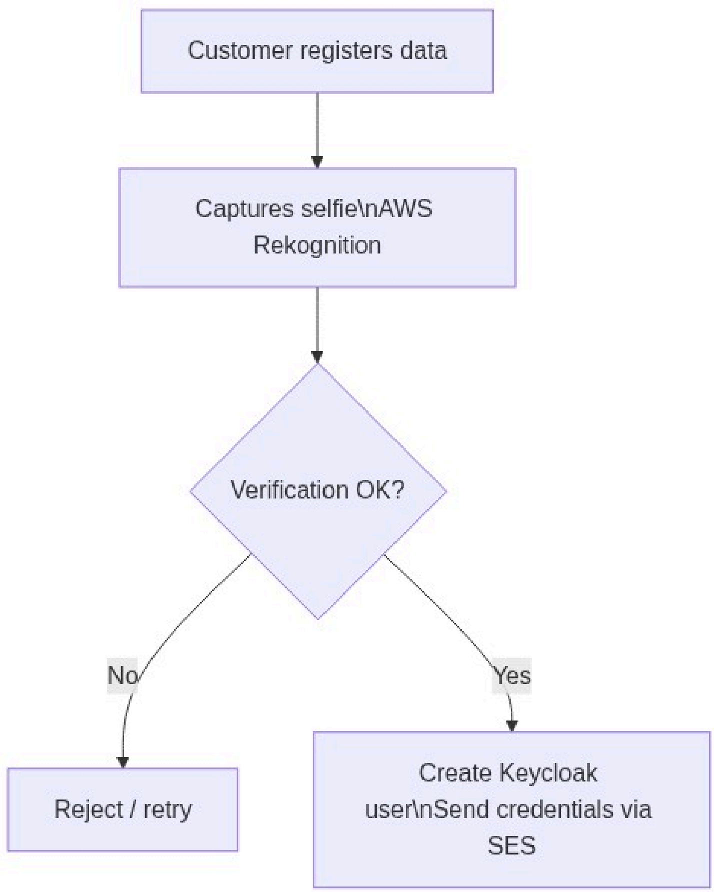
Justification 1: RFC 9700 / OAuth 2.1 removes Implicit Flow due to token leakage risk in URLs and logs. Authorization Code + PKCE mitigates this without requiring a `client_secret` in public apps.

Justification 2: RFC 8252 requires PKCE for mobile apps because they cannot securely store secrets. PKCE uses a device-generated `code_verifier`, making an intercepted authorization code useless.

Post-Onboarding Authentication Methods

Method	Platform	Implementation
Username + password	Web + Mobile	Native Keycloak
Fingerprint	Mobile	Flutter LocalAuth + FIDO2 / WebAuthn
OTP via email/SMS	Web + Mobile	Keycloak OTP plugin
Facial recognition	Mobile	AWS Rekognition + Keycloak plugin

6. Onboarding Architecture — Facial Recognition



Recommended tool: AWS Rekognition — managed service, 99.9% SLA, no internal ML team required. Evaluated alternative: Azure Face API (similar capability, but BP already operates on AWS).

Justification 1: Using a managed service reduces time-to-market and eliminates model training complexity. Rekognition has >99.5% accuracy for 1:1 verification.

Justification 2: Separating Onboarding as an independent microservice allows scaling registration without affecting transactional service availability.

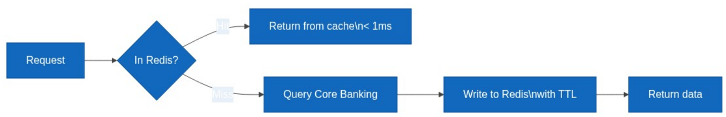
---

7. Cache Pattern — Cache-Aside

Chosen pattern: Cache-Aside (Lazy Loading)

Justification 1: Cache-Aside gives explicit control over what is cached. For sensitive banking data (balances, movements), selective invalidation is critical. Write-Through caches all writes including data that may never be read.

Justification 2: Core Banking is an external system we do not control. Cache-Aside is the only applicable pattern when the data source is a legacy external system with no native write-through support.

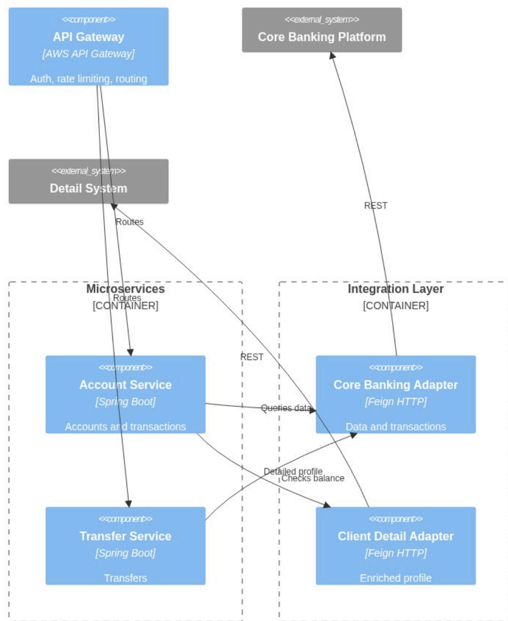


Cached data	TTL	Reason
Customer profile	15 min	Changes infrequently
Products / accounts	5 min	May change due to new contracts
Recent movements	2 min	High change frequency
Session token	Token duration	Fast validation without DB hit

---

8. Integration Layer — API Gateway + Adapters

Integration Layer — API Gateway and Adapters



Justification 1: The API Gateway centralizes authentication, rate limiting and logging. Without it, each microservice would need to implement these cross-cutting concerns, violating DRY.

Justification 2: The Integration Layer with adapters per external system (Adapter Pattern) isolates legacy contracts. If Core Banking changes its API, only the adapter is updated, not the business services.

Defined Services

Service	Endpoint	Source
Basic data	`GET /accounts/{id}`	Core Banking
Movements	`GET /accounts/{id}/movements`	Core Banking
Own transfer	`POST /transfers/own`	Internal + Core
Interbank transfer	`POST /transfers/interbank`	Core + Payment Network
Customer detail	`GET /clients/{id}/profile`	Detail System
Notifications	`POST /notifications` (async)	EventBridge

9. Notification Architecture

Channel 1 — Email (AWS SES): Transaction alerts, transfer confirmations, password recovery.

Channel 2 — SMS (AWS SNS): OTP for MFA, security alerts, critical confirmations.

Pattern: Event-Driven. Services publish events to EventBridge. The Notification Service subscribes and dispatches to the appropriate channel based on customer preferences.

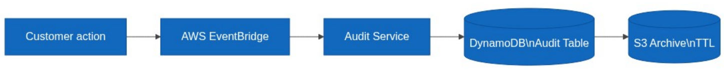
Justification: Decoupling via EventBridge avoids direct dependency between Transfer Service and Notification Service. Using AWS SES + SNS as managed services guarantees 99.9% SLA and complies with financial communication regulations.

10. Audit Architecture

Database: AWS DynamoDB (append-only, no UPDATE or DELETE)

Justification 1: DynamoDB with IAM policy denying UpdateItem and DeleteItem guarantees audit record immutability, complying with SOX and banking regulations.

Justification 2: DynamoDB scales limitlessly, supports native TTL for automatic archival, and encrypts at rest by default — essential for long-term regulatory logs.

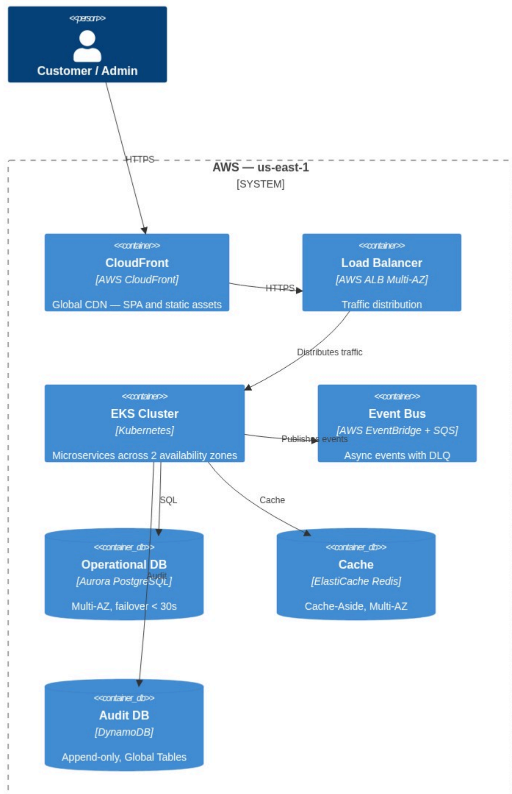


Audit record schema:

Field	Type	Description
`auditId`	String (PK)	UUID v4
`clientId`	String (SK)	Customer ID
`timestamp`	ISO 8601	UTC date and time
`action`	String	LOGIN, TRANSFER, QUERY...
`sourceIP`	String	Source IP
`channel`	String	WEB / MOBILE
`result`	String	SUCCESS / FAILURE

11. Cloud Infrastructure — AWS

AWS Infrastructure — BP Internet Banking



AWS Services Used

Service	Use	Justification
EKS (Kubernetes)	Microservice orchestration	Portability, auto-healing, HPA
Aurora PostgreSQL	Operational database	ACID, Multi-AZ, automatic failover < 30s
DynamoDB	Append-only audit	Unlimited scale, immutability via IAM, native TTL
ElastiCache Redis	Cache-Aside	Sub-millisecond latency, Multi-AZ
API Gateway	Entry gateway	Rate limiting, JWT validation, WAF
EventBridge + SQS	Event bus + DLQ	Decoupling, retries, fault tolerance
CloudFront	CDN for SPA	Low global latency, asset caching
Rekognition / SES / SNS	Biometrics + notifications	Managed services, 99.9% SLA
WAF + Shield	Perimeter security	DDoS protection, OWASP Top 10
CloudWatch + X-Ray	Observability	Metrics, logs, distributed tracing

Service	Use	Justification
Secrets Manager	Secret management	Automatic rotation, no hardcoding

12. High Availability, Fault Tolerance and Disaster Recovery

High Availability

- **Multi-AZ:** All services deployed across minimum 2 availability zones
- **Auto Scaling:** Kubernetes HPA adjusts replicas based on CPU/RPS
- **Circuit Breaker:** Resilience4j prevents failure cascade to external systems
- **Aurora Multi-AZ:** Automatic failover in < 30 seconds

Fault Tolerance

- **SQS Dead Letter Queue:** Failed events after 3 retries go to DLQ
- **Idempotency Keys:** Transfers include `idempotencyKey` to prevent duplicates
- **Graceful Degradation:** If Detail System fails, Account Service returns Core data
- **Timeout + Retry:** Exponential backoff on all HTTP clients

Disaster Recovery

RTO	RPO	Strategy
< 1 hour	< 5 min	Aurora Global Database — replica in secondary region
< 4 hours	< 1 hour	DynamoDB Global Tables replicated in us-west-2
< 15 min	0 (events)	EventBridge replay of archived events

13. Security

Layer	Control	Technology
Perimeter	DDoS, OWASP Top 10	AWS WAF + Shield Advanced
Transport	Encryption in transit	TLS 1.3 on all channels
Authentication	JWT + PKCE + MFA	Keycloak
Authorization	RBAC by role and channel	Spring Security
Data at rest	AES-256	AWS KMS on Aurora, DynamoDB, S3
Secrets	No hardcoding	AWS Secrets Manager with rotation
Containers	Image scanning	Amazon ECR Image Scanning

Regulatory Compliance

Regulation	Application
PCI DSS v4	Card data — encryption, logging, network segmentation
SOX	Immutable audit trail, financial access controls
PSD2 / Open Banking	Strong Customer Authentication (SCA) — mandatory 2FA

Regulation	Application
Data Protection Law	Consent, right to erasure, data minimization
ISO 27001	ISMS — security policies, risk management

14. Monitoring and Operational Excellence

Pillar	Tool	What it measures
Metrics	CloudWatch + Prometheus	CPU, memory, RPS, p95/p99 latency
Logs	CloudWatch Logs	Request traces, errors, audit
Traces	AWS X-Ray	Distributed tracing across microservices
Alerts	CloudWatch Alarms	Anomalies, errors, SLA breach
Dashboards	Grafana	Unified system health view

Production KPIs

Metric	Target
Availability	99.95% monthly
p95 latency (queries)	< 300ms
p95 latency (transfers)	< 2 seconds
Error rate	< 0.1%
Aurora failover	< 30 seconds

15. Architectural Decisions (ADR)

#	Decision	Chosen	Alternative	Justification
1	Architecture	Microservices	Monolith	Independent scalability per service
2	Orchestration	EKS (Kubernetes)	ECS Fargate	Portability, HPA, native self-healing
3	Operational DB	Aurora PostgreSQL	RDS MySQL	ACID, failover < 30s, read replicas
4	Audit DB	DynamoDB	PostgreSQL	Unlimited scale, immutability via IAM
5	Cache	Redis (ElastiCache)	Memcached	Persistence, pub/sub for invalidation
6	Messaging	EventBridge	Kafka	Less overhead, native AWS, declarative routing
7	Auth flow	Authorization Code + PKCE	Implicit Flow	RFC 8252 / OAuth 2.1 — standard for public apps
8	IdP	Keycloak	Custom	Existing corporate product, OAuth 2.0 / OIDC
9	Biometrics	AWS Rekognition	Azure Face API	BP on AWS — lower latency, no cross-cloud egress

#	Decision	Chosen	Alternative	Justification
10	Mobile	Flutter	React Native	Single codebase, native biometric API
11	SPA	React + TypeScript	Angular	Flexibility, TypeScript type safety
12	Cache pattern	Cache-Aside	Write-Through	External Core Banking; explicit invalidation control
13	Notifications	EventBridge + SES + SNS	Twilio	Native AWS, 99.9% SLA, lower cost
14	Gateway	AWS API Gateway	Kong	Less overhead, WAF integration, Lambda authorizer

---

Architecture designed under the C4 model — Context · Container · Component